Economics S5040: Dynamic Stochastic Modeling
Lecture 14 — Computing Optimal Consumption and Saving

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1 The Computational Question

The cake-eating Bellman equation is

$$V(x) = \max_{0 \le c \le x} \{u(c) + \beta V(x - c)\}.$$

For CRRA we know the exact policy,

$$c^*(x) = (1 - \beta^{1/\rho})x.$$

Today we pretend we do not know it and recover the solution numerically:

solve a known problem before trusting the method on an unknown one.

2 The Bellman Operator

For any candidate function v, define

$$\boxed{(Tv)(x) = \max_{0 \le c \le x} \{u(c) + \beta v(x - c)\}.}$$

The value function is a fixed point:

$$\boxed{V = TV.}$$

Value function iteration applies

$$v_{n+1} = Tv_n$$

until successive functions are nearly identical.

3 Discretizing the State

Theory has a continuous state x. Numerically choose

$$x_1, \ldots, x_N.$$

Store $v(x_i)$ as an array. A finer grid generally improves accuracy but costs more computation:

accuracy versus computational cost.

4 Keep the Control Continuous

We need not put consumption on the same grid. At state x_i, solve

$$\max_{0<c\leq x_i} \{u(c) + \beta\widehat{v}(x_i - c)\},$$

where $\widehat{v}$ interpolates the current approximation.

If $x_i - c$ lies between two grid points, linear interpolation supplies the continuation value. This is how a finite array represents a function on a continuous state space.

5 One Bellman Update

At every x_i:

1. choose feasible c;
2. calculate $u(c)$;
3. compute $x' = x_i - c$;
4. interpolate $v(x')$;
5. calculate $u(c) + \beta v(x')$;
6. maximize over c.

Store both the maximum and the maximizing consumption. The former updates the value function; the latter is the greedy policy.

6 Algorithm

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choose x_grid
choose initial v

repeat:
    for x in x_grid:
        maximize u(c) + beta * v(x-c)
        store new value and maximizing c
    error = max(abs(new_v - v))
    v = new_v
until error < tolerance
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Possible initial guesses include $v_0(x) = 0$ or $v_0(x) = u(x)$. The initial guess affects speed, not the economic solution.

7 Stopping Rule

Use the sup norm

$$\|v_{n+1} - v_n\|_\infty = \max_i |v_{n+1}(x_i) - v_n(x_i)|.$$

Stop when

$$\boxed{\|v_{n+1} - v_n\|_\infty < \varepsilon.}$$

8 Recovering and Checking the Policy

After convergence,

$$c^*(x_i) = \arg\max_c \{u(c) + \beta V(x_i - c)\}.$$

Saving is

$$a^*(x_i) = x_i - c^*(x_i).$$

Compare the numerical policy with

$$c_{\text{exact}}(x) = (1 - \beta^{1/\rho})x$$

and calculate

$$\boxed{\max_i |c_{\text{numerical}}(x_i) - c_{\text{exact}}(x_i)|.}$$

Sources of error include grid spacing, interpolation, optimization tolerance, Bellman tolerance, and boundary treatment.

9 Economic Diagnostics

Numerical output should obey economic logic. Since

$$\frac{c^*(x)}{x} = 1 - \beta^{1/\rho},$$

raising β should reduce the consumption share and raise saving. If the computer says otherwise, investigate the computation before inventing an economic story.

10 Productive Saving

Cake eating has

$$x' = x - c.$$

If saving is productive,

$$\boxed{x' = f(x - c).}$$

Then

$$\boxed{V(x) = \max_{0 \le c \le x} \{u(c) + \beta V(f(x - c))\}.}$$

The dynamic-programming structure is unchanged; only the transition changes.

11 Adding Uncertainty

Suppose

$$\boxed{x' = f(x - c)\xi'}$$

for a positive random shock ξ'. The Bellman equation becomes

$$\boxed{V(x) = \max_{0 \le c \le x} \left\{u(c) + \beta E[V(f(x - c)\xi')]\right\}.}$$

The continuation value is now an expected continuation value.

If ξ' takes values ξ_j with probabilities π_j,

$$E[V(f(a)\xi')] = \sum_j \pi_j V(f(a)\xi_j).$$

For a continuous shock, Monte Carlo gives

$$E[V(f(a)\xi')] \approx \frac{1}{M}\sum_{j=1}^{M} V(f(a)\xi_j).$$

Thus Week 3's probability and Monte Carlo tools now sit inside a Bellman equation.

12 The Stochastic Euler Equation

Let $a_t = x_t - c_t$. Since

$$x_{t+1} = f(a_t)\xi_{t+1},$$

the marginal return to one additional unit saved is $f'(a_t)\xi_{t+1}$. Optimality implies

$$\boxed{u'(c_t) = \beta E_t\left[u'(c_{t+1})f'(a_t)\xi_{t+1}\right].}$$

Marginal utility sacrificed today equals expected discounted marginal utility generated tomorrow.

13 A Unifying View

Model	Transition
Cake eating	$x' = x - c$
Productive saving	$x' = f(x - c)$
Stochastic saving	$x' = f(x - c)\xi'$

The state-control-reward-continuation-value structure remains the same.

14 Back to Solow

Solow imposed

$$k_{t+1} = sAk_t^{\alpha} + (1-\delta)k_t.$$

Here saving comes from a policy:

$$\boxed{a^*(x) = x - c^*(x).}$$

We have moved from

$$\boxed{\text{saving as a parameter}} \quad \text{to} \quad \boxed{\text{saving as an optimal policy.}}$$

15 Looking Ahead

Real households receive income and use assets to transfer resources through time. A natural next budget constraint is

$$a_{t+1} = Ra_t + y_t - c_t.$$

With uncertain income, saving also provides self-insurance. This leads toward the income fluctuation problem and precautionary saving.

16 What to Remember

1. $V = TV$ and value function iteration repeatedly applies T.
2. A grid represents a continuous state numerically.
3. Interpolation evaluates continuation values between grid points.
4. Numerical optimization recovers the policy.
5. The exact cake-eating solution provides an accuracy check.
6. Productive saving changes the transition to $x' = f(x - c)$.
7. Uncertainty turns continuation value into expected continuation value.
8. Monte Carlo can approximate that expectation.
9. The stochastic Euler equation equates marginal utility today with the expected discounted marginal return to saving.